

Applications of the InVEST Habitat Quality model: A literature review of methodological trends and model integration, with implications for human-wildlife interactions

Stoyan Nedkov^{1,2}, Katrin Parvanova¹, Sofia Yakovleva³, Ina Vasileva³, Marta Paunova¹

¹ National Institute of Geophysics Geodesy and Geography – Bulgarian Academy of Sciences, Akad. G. Bonchev str. Bl. 3, 1113, Sofia, Bulgaria

² South-West University “Neofit Rilski”, 66 Ivan Mihaylov str. 2700, Blagoevgrad, Bulgaria

³ Sofia University “St. Kliment Ohridski”, 15 Tsar Osvoboditel Blvd. 1504, Sofia, Bulgaria

Corresponding author: Stoyan Nedkov (snedkov@abv.bg)

Abstract

The Integrated Valuation of Ecosystem Services and Trade-offs Habitat Quality (InVEST HQ) model offers a spatially explicit approach for assessing habitat quality utilizing relatively simple, readily available data and an intuitive interface. In this paper, we provide a literature review of the InVEST HQ application based on 61 scientific papers, which have been systematized, analysed, and interpreted using a detailed and structured original template. We identify the main trends and gaps in the model's applications, as well as its readiness for integration into studies on human-wildlife interactions (HWI). The first trend concerns the application of the model for assessing habitat quality to address spatial discrepancies between nature conservation and other human activities. The second one shows that, although InVEST HQ is often considered more suitable for assessing biodiversity in general, our review identifies a parallel trend of using the model in species-specific studies. The third trend reveals the application of the model in combination with other models and methods, which can provide more robust and ecologically meaningful results, compensating for its reliance on expert-based parameters. We further examine the extent to which InVEST HQ applications relate to HWI, showing that although the applicability is currently limited and mostly indirect, the model's core strengths make it a promising tool for future HWI-oriented research. These results highlight the increasing popularity of the model as a spatially explicit tool for assessing habitat quality under a wide range of contexts, and point to its widespread integration with complementary tools as a core spatial framework that requires additional modelling layers to capture ecological complexity. This is in line with current trends in integrated ecosystem services modelling, which is necessary due to the multiple trade-offs among different services and the associated challenges that ecosystem assessments aim to resolve.

Key words: Habitat maintenance, literature review, models integration, nature conservation, spatially-explicit modelling



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Introduction

Human-wildlife interactions (HWI), in the form of coexistence and conflicts, have become increasingly frequent as human populations expand and natural

habitats are altered. The study of these interactions has become a key paradigm in European biodiversity protection and conservation management (König et al. 2020). Sustainable concepts of sharing space that take into account the needs of wildlife species, stakeholders' interests, and regional landscape conditions are much needed. Wildlife conservation goals should not be compromised. Instead, alternative livelihoods, such as ecotourism, beekeeping, or livestock fattening, should be recommended (Temesgen et al. 2025). However, conflicts between human needs and wildlife conservation efforts remain predominant in HWI (Khan et al. 2025). The studies on the spatial aspects of these conflicts are very important for gathering information to be used for risk assessment and predictions that could help to find alternative pathways for HWI that promote sustainable coexistence. Spatially explicit models that assess habitat quality, such as the Integrated Valuation of Ecosystem Services and Tradeoffs (InVEST) Habitat Quality (HQ) model discussed in this review, provide one such tool for generating this information.

In this context, the ecosystem services (ES) concept offers a robust and policy-relevant approach for addressing HWI by explicitly linking biodiversity conservation with human well-being and risk management (Ceaşu et al. 2019). Integrating ES into the spatial analysis of HWI enables the identification of conflict hotspots, trade-offs, and synergies between conservation objectives and human activities, thereby strengthening risk assessment and predictive modelling. Such spatially explicit GIS-based approaches are particularly relevant in European landscapes, where conservation and land-use planning are guided by initiatives such as the Mapping and Assessment of Ecosystems and their Services (MAES) framework (Maes et al. 2012; Burkhard et al. 2018). ES modelling enables the spatial quantification and mapping of ecosystem service supply, demand, and flows (Crossman et al. 2013), thereby revealing trade-offs and synergies of different activities that constitute HWI (Bennett et al. 2009). By integrating ecological data with land-use, socio-economic, and biophysical variables, ES models support scenario-based analyses that can predict how changes in wildlife populations, habitat configuration, or management interventions affect both ecosystem service provision and conflict dynamics (Nelson et al. 2009; Guerry et al. 2015). Such spatially explicit modelling approaches are particularly valuable for risk-oriented decision-making, as they help identify priority areas where conservation actions can simultaneously reduce human-wildlife conflict and enhance ES (Morzillo et al. 2014; Nyhus 2016; Balasubramaniam et al. 2021).

Habitat maintenance represents a critical regulating and supporting ecosystem service that underpins both biodiversity conservation and the stability of socio-ecological systems (Hatziiordanou et al. 2019; Borisova et al. 2023). Habitat maintenance refers to the capacity of ecosystems to sustain viable wildlife populations through the provision of suitable habitat structure, connectivity, and ecological processes. The assessment of habitat quality is among the most important activities in HWI studies. It necessitates long-term monitoring data and high-quality experts for producing appropriate results. However, the decision-makers often need fast and easy-to-use tools and approaches to meet the challenges of increasing human-wildlife conflicts. The InVEST HQ model is an appropriate tool to estimate the impact of different scenarios of land use and land cover (LULC) change or policies on habitats for biodiversity conservation with relatively low data requirements (Terrado et al. 2015).

There is a growing need for applied, spatially explicit approaches that can capture the complexity of human-wildlife interactions while remaining accessible to decision-makers. Despite advances in ES modelling and habitat assessment, a key challenge remains the development of methods that effectively translate spatial patterns of human-wildlife interactions into actionable information for risk assessment and prediction (McLain et al. 2013; Naha et al. 2019; González-Crespo et al. 2023). In particular, relatively simple and user-friendly tools capable of integrating spatial data on land use, habitat quality, and human pressures are needed to support timely management and planning decisions. Therefore, it is important to define what spatial patterns characterize human-wildlife interactions, and to what extent these patterns can be modelled for the purposes of risk assessment and prediction using relatively simple and user-friendly tools. Habitat quality is one of the most important characteristics that highly depends on spatial pattern, and its modelling can ensure valuable data are used for risk assessment and prediction. The main output of the InVEST HQ model provides such data in the form of spatially explicit results by using relatively simple, easily available data and an easy-to-use approach. Therefore, a study on its applicability for revealing the spatial pattern of HWI is warranted.

The first mention of InVEST HQ in the scientific literature appears in the work of Nelson et al. (2009), which introduced and demonstrated the InVEST modelling framework, including the quantification of habitat provision/quality as an ecosystem service. The HQ model appears in the documentation for version 3.0 of InVEST, which dates to 2014. The first recorded application of the InVEST HQ model in a peer-reviewed research paper appears to be the work of Duarte et al. (2016). They used it in combination with a set of GIS procedures to assess the overlap between three different ES (carbon stock, sediment retention, habitat quality). The first paper where the model is used as a single tool is the study that proposes an approach that supports the definition of sustainable development policies for preservation of habitat structures while promoting their exploitation as cultural and landscape assets in an urban planning context (Epifani et al. 2017). The application of InVEST HQ for a particular biological entity is presented for the first time in the work of Sulistyawan et al. (2017), which explores the means of connecting fragmented habitat corridors of endangered mammals to design wildlife corridors through habitat quality assessment. The model appears to be useful for studies in urban landscapes focused on the impact of built-up areas and LULC changes on biodiversity and natural habitat loss (Hack et al. 2020; Tang et al. 2021; Erdoğan 2025), assisting decision-making in multifunctional green infrastructure planning (Dong et al. 2025), and variations of bird diversity in rapidly urbanized regions (Li et al. 2021). Recently, there have been studies on habitat quality of specific animal species, including Tiger (*Panthera tigris* (Linnaeus, 1758)) habitat quality modelling (Louis et al. 2024), impact of village development on the habitat quality of Snub-nosed Monkeys (*Rhinopithecus* Milne-Edwards, 1872 spp.) (Zhu et al. 2022), impact of land-use change on Crocodile Lizards (*Shinisaurus crocodilurus* Ahl, 1830) (Zhang et al. 2022b), and elephant habitat quality modelling (Mukhopadhyay et al. 2025b).

Despite growing attention and more studies using the InVEST HQ model for various purposes ranging from nature conservation to restoration activities and urban planning, there are still knowledge gaps in its application, especially in

studies focused on HWI. First, the rapidly increasing number of publications makes a systematic synthesis of the literature imperative. Existing reviews have addressed the broader InVEST model suite, for example by quantifying the relative frequency of use of its different modules, including HQ, across the global literature (Mukhopadhyay et al. 2025a), but none has specifically consolidated and critically evaluated the methodological choices underlying InVEST HQ applications, such as the design of threats and sensitivity parameters, or their combination with other models and approaches. Moreover, no existing review has systematically examined the extent to which InVEST HQ applications relate to human-wildlife interactions. This review therefore addresses a distinct gap by providing the first template-based synthesis of InVEST HQ methodological choices across general and species-specific applications, a structured classification of the models and approaches with which it has been combined, and an explicit assessment of the model's relevance to HWI research.

In response to these gaps, the main objective of this review is to synthesize the existing literature on the application of the InVEST HQ, systematically assessing methodological trends and patterns of model integration, and evaluating the model's readiness for application in studies of HWI. Specifically, we aim (i) to consolidate and critically assess the rapidly expanding body of InVEST HQ applications in order to provide a structured overview of current research trends in habitat quality and related ES; (ii) to clarify how the model has been applied in different studies, with a focus on species-specific contexts, evaluating the extent to which species- or group-level ecological requirements are incorporated and to identify common approaches and limitations, including their combination with other models and methods; and (iii) to examine in detail the types of input data used and the methods by which threats and species sensitivity parameters are defined and adjusted, including their combination with other models and methods in HWI studies.

Materials and methods

InVEST is a suite of open-source software models developed by the Natural Capital Project, which is a partnership involving Stanford University, the University of Minnesota, The Nature Conservancy, and WWF (Natural Capital Alliance 2025). Its main objective is to provide appropriate means for mapping and assessment of ES and help decision-makers understand how changes in ecosystems affect both human well-being and biodiversity (Tallis and Polasky 2009). Currently, it incorporates 20 models focused on different ES, and InVEST HQ is one of them. To fulfil the main objective of the study, we arranged a literature review that was conducted in three main stages: 1) literature search; 2) papers review; and 3) data analysis. Each of them is described in the following subsections.

Literature search

The first step of the review was to identify all available peer-reviewed publications in the Scopus and Web of Science databases that applied the InVEST HQ model and addressed the study's objectives. The search string was (InVEST) AND (habitat quality) AND (model OR modelling OR tool). The search resulted in the finding of 75 papers, which have been further assessed against two main

criteria. The first was to have a real application of the model. In some publications, the model can be mentioned in a particular context but not applied as a tool for assessing habitat quality. Therefore, we checked each paper to determine whether the application of the model is properly documented and the results from the application are presented. The second criterion was the language of the paper. Although most of the papers were in English, some of them were in Chinese or Korean. Since we did not have the necessary skills to read text in these languages, the respective papers were removed. Thus, the final number of papers to be reviewed at the next stage was reduced to 61 (Suppl. material 1).

Papers review

Review template

To organize the review process for systematizing and checking the content of the reviewed papers, a specialized database with a standard nomenclature was constructed (Ochoa and Urbina-Cardona 2017; Perevochtchikova et al. 2019; Nedkov et al. 2022). It was organized as a template in the form of a structured MS Excel sheet (Suppl. material 2) designed to fulfil the main objectives of the study. The template contains three sheets (Metadata, InHQ1, and InHQ2) and the data entry was organized into six components: 1) General characteristics of the publication; 2) Case study; 3) InVEST HQ application; 4) Practical aspect of the application; 5) Ecosystem services; and 6) Specifics of the InVEST HQ application (see Suppl. material 2: Metadata). The sheet InHQ1 contains a limited number of variables and was used for the initial check while the sheet InHQ2 contains the full number of variables for the in-depth review of the papers.

Data entries and the main components of the review template

The preferable method for data entries during the papers review was the binary numerical system; in the other cases, the inputs were in the form of text (Nedkov et al. 2022, 2024). The data entries for each paper were stored in a single row to facilitate the data processing and statistical analyses. The papers were distributed between the five members of the team and everyone followed a uniform approach to extract the necessary information for each of the components. The content and the procedure for data entry per each component are given below.

The first component on the general characteristics of the publication contained bibliographic data about the authors, the journal and its indexing information, the year of publication, the language, and the DOI. The second was focused on the case study area where the model has been applied. Apart from the standard characteristics such as case study name and country, we also included the scale and type of the study area. The latter would enable us to define the appropriate scales of the model application (from global to local) and the preferred type of spatial unit (administrative, watershed, protected area, natural unit, etc.). The third component was designed to define the main aspects of the model's application. The first aspect was whether the model was applied for habitat quality in general or if it was species specific. The latter was divided into plant and animal species, as well, each of them was specified according to the particular entity (specific species, group of species, or in general for animal or plant

biodiversity). The second aspect concerned the methodological approach, more specifically, whether the InVEST HQ was applied alone or in combination with other methods. The latter were divided into two categories: models and other methods. The fourth included characteristics concerning the decision-making and policy aspects of the model's application. The main categories were defined as nature conservation, urban planning, public sector management, restoration activities, and others. Furthermore, a multiple purposes option was added for complex studies that cover more than one of the predefined categories. The fifth component contained data about ES assessment in the study. InVEST is a model for mapping and assessing ES, and it is important to reveal how they are addressed in the studies. The sixth part was designed to reveal the specifics of the model's application with respect to its main inputs and their characteristics. InVEST HQ requires three main inputs (LULC, threat table, and sensitivity table), which were correspondingly incorporated in the template. Special attention was put on the model's advantages, disadvantages, and limitations.

After the review of the papers, all data entries from the five reviewers were integrated into a single sheet and checked for errors and consistency. For some components, additional fields were added to generate new data based on the information gathered from the papers. For instance, the information on threats, which was grouped into four main categories (urban, agriculture, roads, and other), were reclassified into seven categories (see Fig. 5). Thus, a database for use in the analyses was created.

Data analyses

Classification schemes

Prior to the analyses, several of the extracted variables required additional classification into analytical categories, beyond the direct data-entry fields described above. These schemes were developed iteratively during the review process and applied consistently across the dataset (Table 1).

Models and approaches combined with InVEST HQ were classified into two categories: models and other approaches. The models were further subdivided into groups according to their functional role into: (i) statistical; (ii) landscape ecology; (iii) GIS/spatial analysis; and (iv) other models. The other approaches were subdivided into: (i) statistical; (ii) landscape ecology; (iii) GIS/spatial analysis; (iv) remote sensing and spectral indices; (v) LULC change analyses; and (vi) other approaches (Table 1).

Table 1. Summary of the analytical framework.

Analysis group	Objective	Variables analysed	Type of analysis
1. General characteristics	Characterize publication trends and spatial distribution	Year, journal, country, spatial scale	Descriptive statistics; full dataset vs. species-specific subset
2. Model application & ES	Characterize methodological application and ES relation	Object of assessment, species group, model/approach combinations, decision-making context, ES category	Descriptive statistics; cross-tabulation; general vs. species-specific comparison
3. Model specifics & HWI	Characterize technical design and relation to HWI	LULC type/resolution, threats, sensitivity classes, suitability values, outputs, HWI category/aspect	Descriptive statistics; cross-analysis against HWI categories

Relation to ES was classified into three categories: (i) papers performing a specific ES assessment; (ii) papers using habitat quality as a proxy for biodiversity and ES; and (iii) papers applying the model as a supporting component without directly addressing ES.

LULC input type was classified into standard (existing, multi-purpose, largely freely available datasets) and user-defined (developed specifically for the study).

Relation to HWI was classified into three categories (direct, indirect, and no relation). Papers with a direct or indirect relation were further grouped into four recurring HWI aspects: (i) wildlife protection under threat; (ii) interaction between a specific human activity and the habitat quality of a given species; (iii) human impact on biodiversity or specific species; and (iv) measures to ensure favourable habitats for wildlife species.

Analytical framework

The data analyses were divided into three groups: corresponding to the review's main objectives (Table 1).

The first group is focused on the general characteristics of the publications. Descriptive statistics were used to characterize publication trends (temporal distribution, journal distribution) and the spatial distribution of case studies (country, spatial scale). These were computed for the full dataset and, separately, for the subset of species-specific papers, to allow comparison between the two.

The second group deals with the model application and their relation to ES. This group combined descriptive and comparative analyses of the distribution of studies by object of assessment (general vs. species-specific) and taxonomic group, patterns of combination between InVEST HQ and other models/approaches (including how these patterns differ between general and species-specific studies), the distribution of studies across decision-making and policy contexts, and the distribution of studies across the three ES-relation categories (see Classification schemes, above).

The third group is focused on model specifics and relation to HWI. This group analyzed the frequency and design of model inputs (LULC type and resolution, number and type of threats, number of LULC classes, and suitability value type in the sensitivity table) and outputs (habitat quality vs. habitat degradation grids), comparing general and species-specific subsets. These technical characteristics were then cross-analyzed against the HWI relation categories and the four HWI aspects in order to examine how model design choices relate to the different types of HWI addressed in the reviewed literature.

Results

Characteristics of the publications

In total, 61 papers were selected for review, covering the period from 2016 to 2025. The number of papers per year increased overall, though with some fluctuations, rising from 1 in 2016 to 13 in 2025 (Fig. 1A). The general trend indicates a clear upward trajectory in research output across the entire period, despite intermittent declines in certain years (e.g., 2021 and 2023). The majority of the papers (approximately 82%) were published after 2019, marking a phase of

accelerated growth in publication activity. During this period, the annual output increased from fewer than 5 papers per year (2016–2018) to consistently higher levels, reaching between 9 and 13 papers per year in the later stages (2020–2025). This shift suggests a growing research interest in the InVEST HQ model.

The publications were distributed across a wide range of scientific journals, indicating a high level of dispersion in publication outlets. In total, the papers appeared in 36 different journals; however, only a limited number of journals accounted for more than one publication. The highest number of papers were published in “Sustainability” (6), followed by “Ecological Indicators”, “Land”, and “Remote Sensing” (5 each), and “Global Ecology and Conservation” (4) (Fig. 1B).

Most of the remaining journals hosted only a small number of publications. Specifically, six journals published two papers each (“Journal of Cleaner Production”, “Land Degradation and Development”, “Land Use Policy”, “Lecture Notes in Computer Science”, “Polish Journal of Environmental Studies”, and “Frontiers in Environmental Science”), while the largest share of journals (29) contained only a single publication.

Regarding the application of the InVEST HQ model, we looked at the scale at which it was applied and found that almost all case studies focused on the local (29) or regional (28) level, with only 4 addressing the topic at a national (2) or subnational (2) scale. This clearly demonstrates that the InVEST HQ model is usually applied at the regional or local scale and is rarely used at a national or subnational level.

The case study countries were distributed across the world, with clear dominance by China, which accounts for 49.2% of all case studies ($n = 30$), followed by Italy ($n = 7$), and Iran ($n = 4$). This is illustrated in the map (Fig. 1D), where country shading and pie chart size are both proportional to the number of case studies per country. A minimum size threshold is applied to single-study countries for cartographic visibility. Three multi-country case studies (Kenya–Tanzania–Zambia; Italy–Portugal; India–Bangladesh) are represented as independent pie chart segments to preserve spatial accuracy. Pie chart segments show the proportional composition of case study area types per country. The remaining countries contribute one to two studies each, indicating that model application is driven by a small number of high-output research contexts rather than a globally distributed evidence base. Within-country thematic diversity is greatest in China, where studies span administrative, urban, natural, basin, and protected area contexts (Fig. 1C).

Model application

The main purpose of the InVEST HQ model is to assess the habitat quality of a given area. Its application in the reviewed papers revealed two distinct cases with regard to the object of assessment. In 43 of the studies, biodiversity in general was considered. In these cases, habitat quality is assessed in a broad sense, predominantly in relation to biodiversity conservation. The second case comprises 18 papers that focus on specific groups of species or individual species. The distribution of these studies according to the examined species is presented in Fig. 2. The results show a clear predominance of studies on animals compared with plants (15 versus 3). Among the animal-focused studies, only one considers animals as a whole, five focus on specific groups

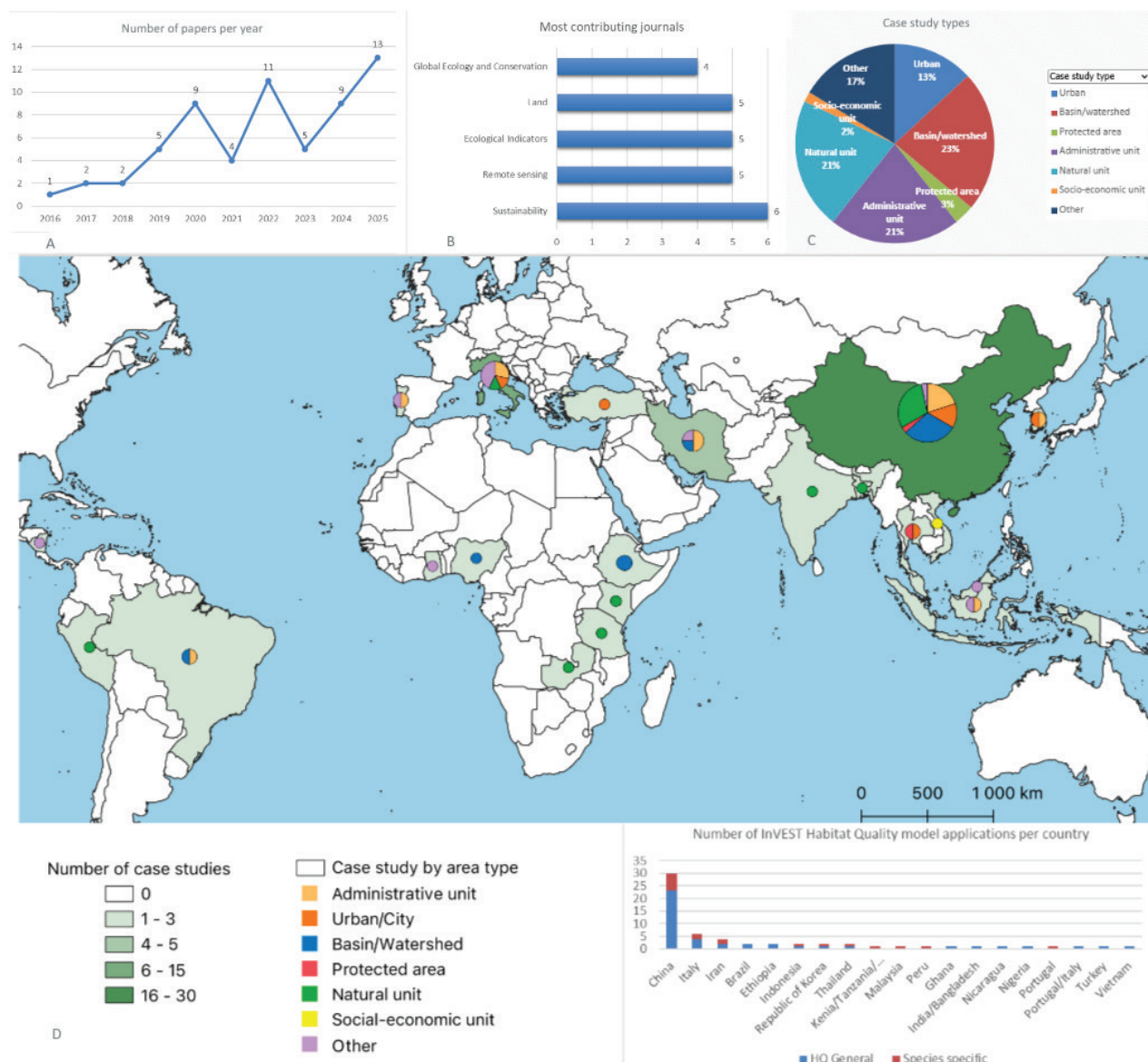


Figure 1. Characteristics of the publications: **A.** Number of publications per year; **B.** Most contributing journals; **C.** Types of case studies; **D.** Map of case studies in the reviewed publications.

(e.g., mammals or birds), and the remaining studies examine habitat quality in relation to individual species (Fig. 2A). Birds are more frequently addressed in studies focusing on animal groups (4 out of 5), whereas mammals are more often the focus in species-specific studies (5 out of 9; Fig. 2C). Six papers focus on a single species (Fig. 2B). These species are tiger, (*Panthera tigris* (Linnaeus, 1758)) snub-nosed monkey (*Rhinopithecus* Milne-Edwards, 1872 spp.), elephant (*Elephas maximus* Linnaeus, 1758 / *Loxodonta* spp.), crocodile lizard (*Shinisaurus crocodilurus* Ahl, 1830) and pancake tortoise (*Malacochersus tornieri* (Siebenrock, 1903)). A special case is the study on *Ericaria amentacea*, as algae represent a distinct biological group. The remaining studies include analyses of multiple species, covering two species (yellow-throated martens (*Martes flavigula* Boddaert, 1785) and leopard cats (*Prionailurus bengalensis* (Kerr, 1792))), three species (Peruvian night monkey (*Aotus miconax* Thomas, 1927), yellow-tailed monkey (*Lagothrix flavicauda*

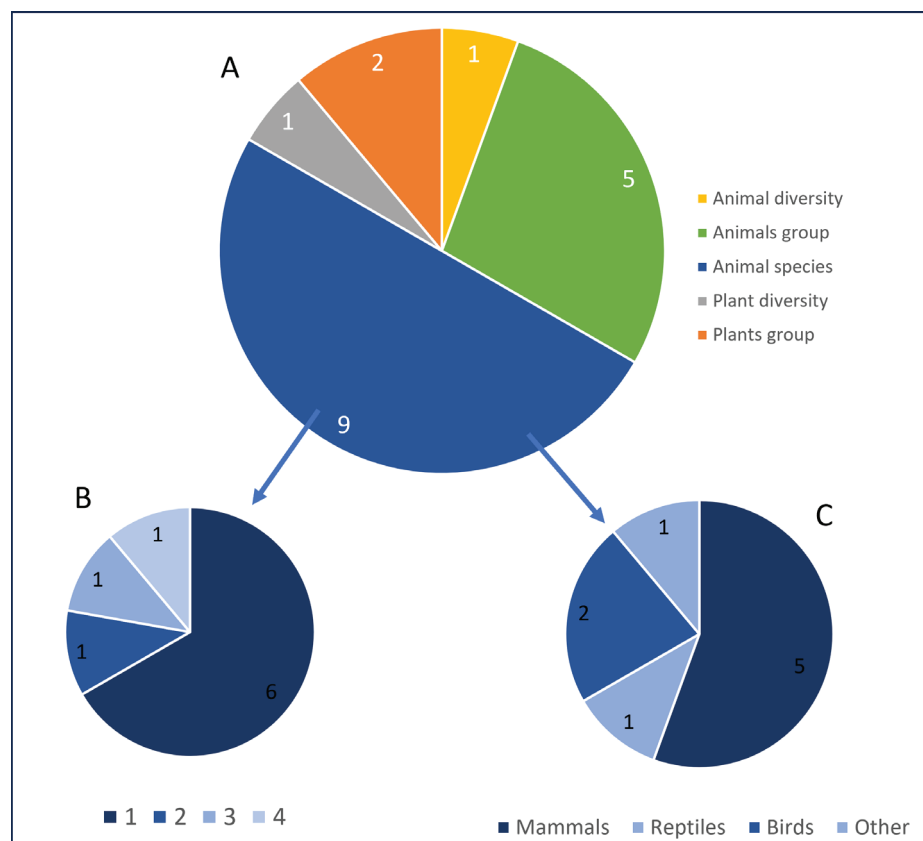


Figure 2. Distribution of species-specific papers: **A.** Distribution of all 18 papers; **B.** Number of animals in papers that study specific animals; **C.** Groups of species in papers that study specific species.

(Humboldt, 1812)) and spectacled bear (*Tremarctos ornatus* (F. Cuvier, 1825))), and four species (Egyptian vulture (*Neophron percnopterus* (Linnaeus, 1758))), eastern imperial eagle (*Aquila heliaca* Savigny, 1809), saker falcon (*Falco cherrug* Gray, 1834) and Eurasian hobby (*Falco subbuteo* Linnaeus, 1758)).

The analysis of the decision-making and policy context revealed an uneven distribution of the reviewed studies across the predefined categories. Nature conservation is the most prominent category, comprising 24 studies, followed by public sector management (6) and urban planning (5), while restoration activities (3) and other applications (1) are less frequently represented. In addition, a considerable number of studies (22) address multiple purposes. Within this group, nature conservation is again the most frequently represented category (22), followed by public sector management (14), urban planning (6), and restoration activities (5). These results highlight the dominant role of nature conservation as a primary application domain, while also demonstrating the model's capacity to support multiple, overlapping decision-making contexts. The diversity of application contexts is further reflected in the variety of analytical tools and models used in combination with InVEST HQ.

The literature review identified 34 cases where models, and 53 cases where other approaches were used in combination with InVEST HQ. The models identified in the reviewed papers were grouped into four categories based on the specifics of their application and functionalities (Table 2). The largest group is landscape ecology models, which can be explained by their close alignment in

Table 2. Number of models and approaches used in the reviewed papers (some models and approaches are used in more than one paper).

Models	Total N	Individual N	Approaches	Total N	Individual N
Statistical	7	5	Statistical	3	3
Landscape ecology	19	12	Landscape ecology	6	4
GIS/Spatial analyses	5	5	GIS/Spatial analyses	20	19
Others	3	3	RS and spectral indices	10	8
			LULC change analyses	5	3
			Others	12	11

terms of functionality and application to the topic of habitat quality. Although a variety of models are used in the papers, some of them, such as Fragstat and MCR (Minimum Cumulative Resistance model), appear in more than one study, with each used in four different papers. Statistical models form the second largest group, with seven papers in total and five individual models. MAXENT and Markov modelling appear in more than one paper. GIS-based spatial analysis modelling tools, such as GWR (geographically weighted regression) and the Gravity model, are used only occasionally, each in a single paper. Three other models, the economic cost model, Expert-Based Models (EXBAS), and the ANN (Artificial Neural Network) model, do not fit the previous groups and were placed in a separate group. The approaches were systematised into similar groups, but as they were more diverse, two additional groups were distinguished. Most of the approaches are unique, as they are developed and proposed for each specific study. The most notable exception is MSPA (Morphological Spatial Pattern Analysis) from the group of landscape ecology approaches, which appears in three different studies. GIS-based spatial analysis approaches are by far the most used, followed by Remote Sensing (RS), and spectral indices. The average number of methods used in the reviewed papers is 2.5, and it is higher in the general papers (2.9) than in species-specific papers (2.2).

The analysis of different combinations between InVEST HQ and the identified groups of models and approaches revealed a particular pattern of distributions, shown in Fig. 3. The share of papers where it is used alone is 25%. These percentages vary between general (23%) and species-specific (28%) papers. In the other papers, InVEST HQ is combined with other models (18%), approaches (28%), and a combination of both models and approaches (30%). Combinations between InVEST HQ and other models from the InVEST toolbox (m2), such as Carbon Storage and Sequestration, Sediment Delivery Ratio, and Seasonal Water Yield, were found only in the fourth group (combination of both models and approaches). These combinations occur in four papers altogether, and all represent the general application of the model. Another combination (m2), which was found only in the fourth group, includes other models from the InVEST toolbox, but in this case, they were combined with statistical models (CA Markov). The opposite situation was observed with combinations m7 and m8, which were found only in the second group (combination of InVEST HQ with other models). The other four combinations between models were distributed evenly between the two groups. The distribution of combinations between approaches showed a similar pattern, as some of them (o3–o7) were found in both groups, while others (o1 and o2) occurred in only one group.

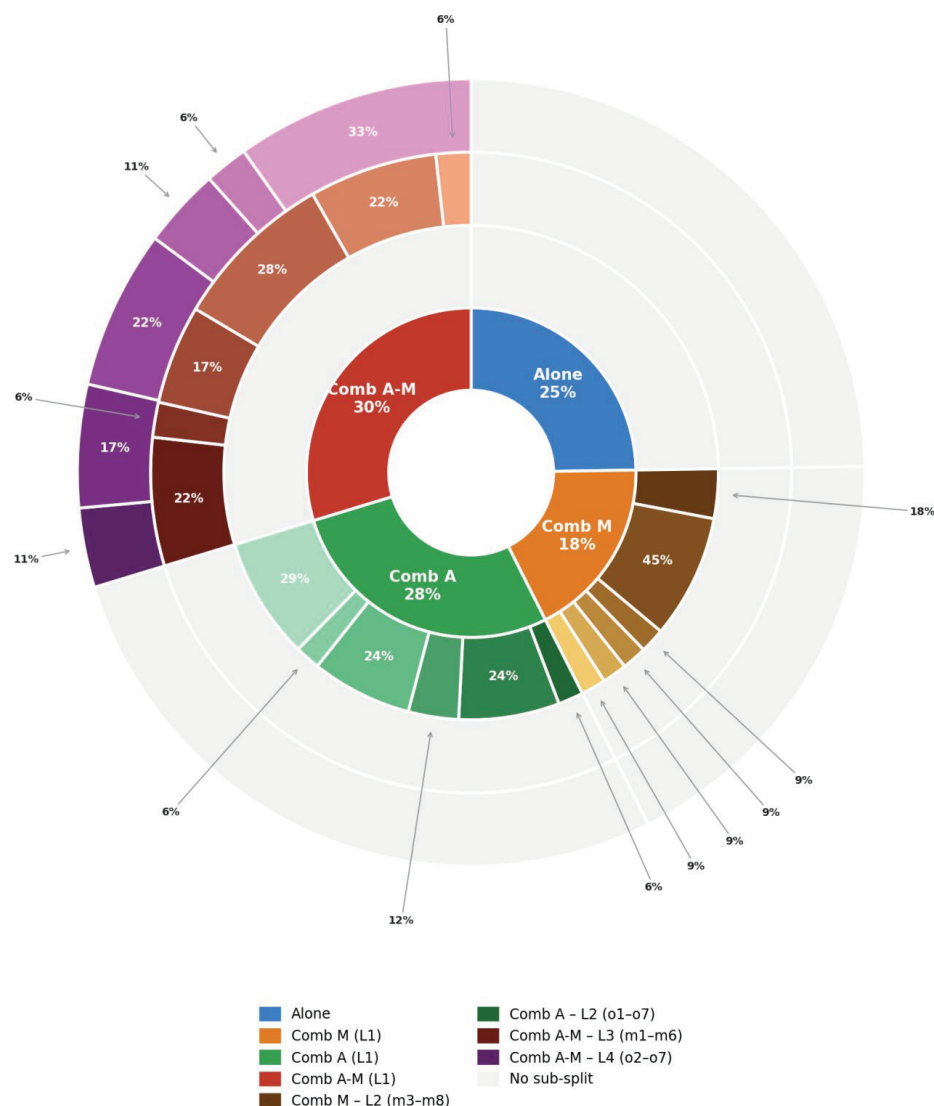


Figure 3. Use of the InVEST HQ in combination with other models and approaches: Comb M – InVEST HQ in combination with other models; Comb A – InVEST HQ in combination with other methods or approaches; Comb A-M – InVEST HQ in combination with both models and approaches; m1 – Combination with other InVEST models; m2 – Combination with other InVEST model and a statistical model; m3 – Combination with statistical models; m4 – Combination with landscape ecology models; m5 – Combination with spatial analysis models; m6 – Combination with other models; m7 – Combination with landscape ecology and statistical models; m8 – Combination with landscape ecology and other models; m9 – Combination with other approaches; o1 – Combination with statistical approaches; o2 – Combination with landscape ecology approaches; o3 – Combination with spatial analysis approaches; o4 – Combination with other approaches; o5 – Combination with RM approaches; o6 – Combination with LULC change analysis approaches; o7 – Combination of approaches.

The analysis of the InVEST HQ application in relation to ES showed that most of the papers (64%) fall into the category where habitat quality is used as a proxy for biodiversity and ES. A total of 12 papers (20%) use the model as a supporting component without directly addressing ES, while only 10 papers (16%) present a specific ES assessment. The number of individual ES examined in these studies ranges from two (in four papers) to eight (in one paper).

The results indicate a clear predominance of regulating services. In addition to habitat quality, eight other regulating services are considered. These include carbon storage and climate regulation (in five papers), flood regulation (three), sediment retention (two), biodiversity maintenance (two), nutrient cycling (two), ecological integrity (one), air purification (one), and pollution purification (one). Provisioning services are less frequently considered, being addressed in four papers and covering water yield, water supply, crop provision, and raw materials. No cultural ES are identified in the reviewed studies.

Model specifics and relation to human-wildlife interactions

The analyses of the model specifics considered how inputs and outputs were presented in the papers and how they were designed in the different studies. The main spatial input is LULC, which appears in most of the papers (57 out of 61). However, four studies use different spatial units, such as habitat and ecosystem types. Habitat types can be considered the most detailed and appropriate for assessing habitat quality, and they were used in three studies (Mashizi and Sharafatmandrad 2019; Gu et al. 2023; Bordoni et al. 2025). Ecosystem types were used in only one study (Meza Mori et al. 2025), but this is one of the most relevant to HWI. Three of these studies belong to the group of species-specific studies, and all of them are relevant to HWI. The LULC types used in the studies are shown in Fig. 4. The majority of the studies (60%) used standard types, and most of these (68%) are based on national LULC datasets. The resolution of the datasets ranged from 2 m to 1,000 m, with the highest proportion (41%) using a resolution of 11 to 30 m (Table 3). A relatively large number (33%) of studies did not specify the resolution of the LULC used.

Almost all reviewed papers, except for three, include appropriate information about the threats considered in the InVEST HQ application. Therefore, the analyses presented here are based on information from 58 papers. The number of threats varies between two and 16 across the papers (Fig. 5A). However, the typical number of threats ranges from three to nine, with higher and lower values being relatively rare. The most common number of threats is six, reported in 13 papers (22% of all papers). The species-specific studies show a similar pattern, except that the most common number of threats is five rather than six.

The threats used in the different studies were grouped into seven categories (urban, agriculture, roads, industry, bare lands, abiotic risks, and biotic risks), and their distribution among the reviewed papers is presented in Fig. 5B. Urban, agricultural, and road threats were the most common in the majority of the papers. Urban areas were considered a threat in 90% of the studies, but

Table 3. LULC resolution in the reviewed papers.

Resolution (m)	N papers	%
<10	4	7
11 to 30	25	41
31 to 100	5	8
101 to 300	3	5
>300	4	7
not specified	20	33

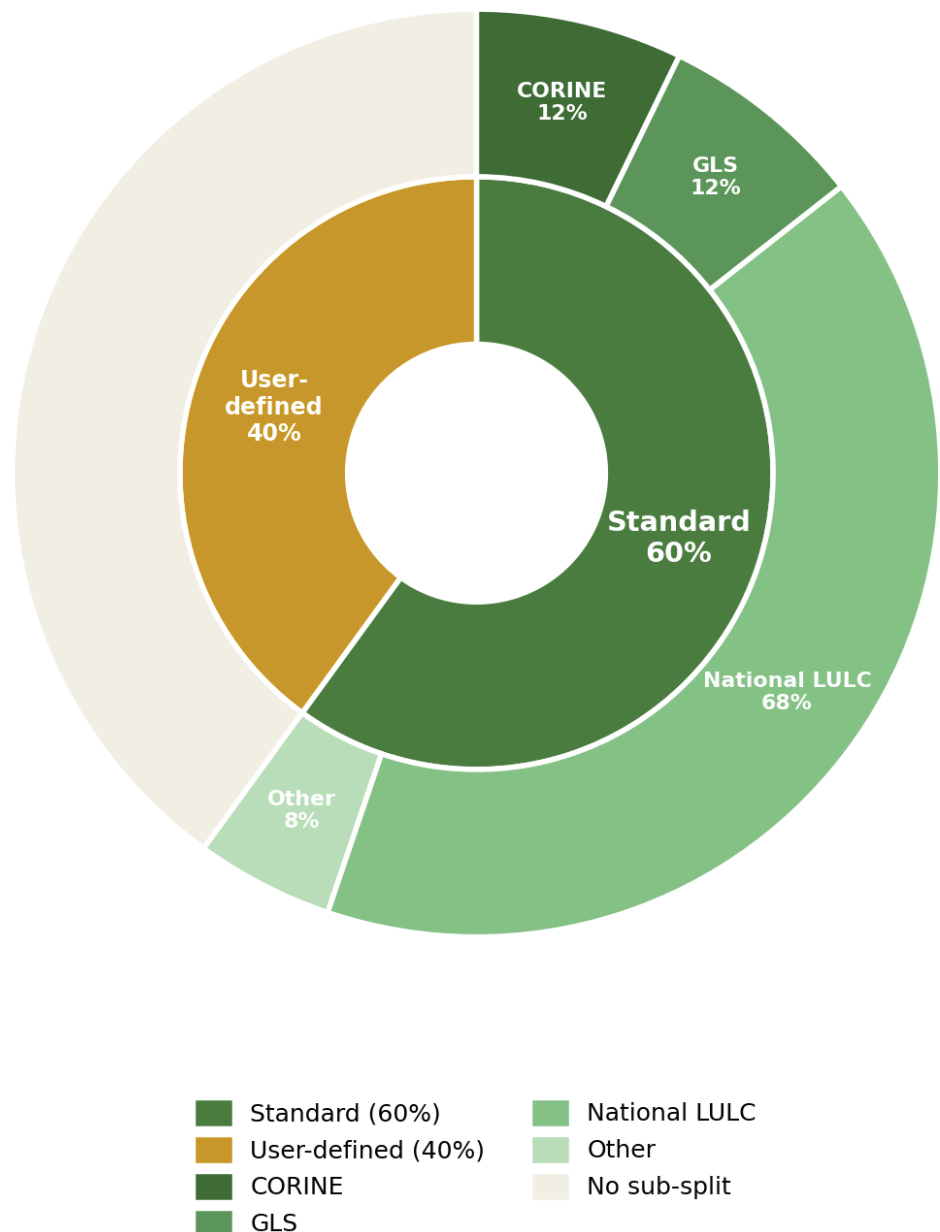


Figure 4. LULC types used in the reviewed papers.

in the species-specific studies the percentage was even higher (94%). Almost half of the studies (54%) considered urban areas as a single threat, but in the rest they were differentiated into distinct categories that varied between two to seven. In most cases (15), they were divided into urban residential and rural residential. Other variants of urban threat included population density (3), recreational or accommodation facilities (3), public utilities (2), community built-up lands (2), and other construction lands (2). A similar pattern was observed for road threats, considered 74% in all studies and 83% in species-specific studies. Roads as a single threat were considered in 40% of the studies, while in the others they were divided into between two to five categories. This division usually included the different road classes (including railways). The share of agriculture as a threat was higher in all papers (84%) and slightly lower (78%) in the species-related papers. It was considered a single threat in most of the studies (78%), while in the others it was divided into two to four categories.

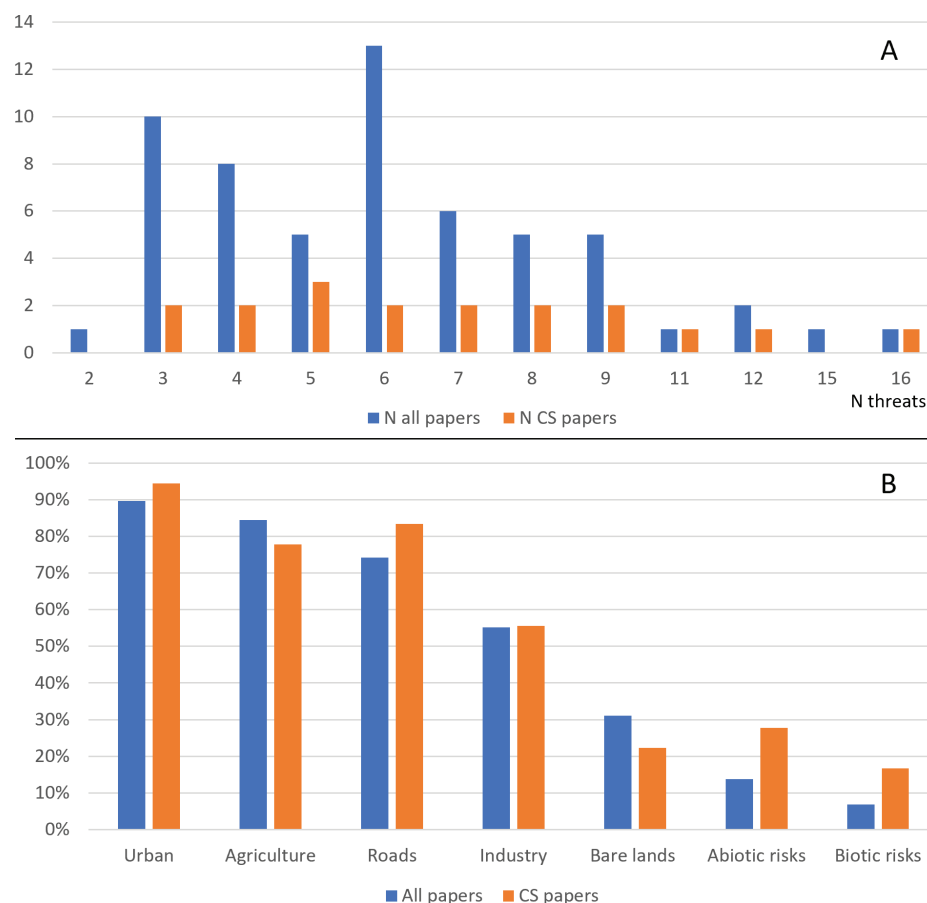


Figure 5. Distribution of threats in the reviewed papers: **A.** Number of threats per paper; **B.** Groups of threats used in the papers.

Threats from industry were taken into account in almost half of the studies, with no difference between the total number of papers and the species-specific ones. Bare land, and abiotic and biotic risks were presented as threats in fewer studies, but there were notable differences between the groups (all studies and species-specific). Bare land was considered a threat in almost one third of the studies, but in the group of species-specific studies it was less represented (22%). In most of the studies, it was considered a single threat. Both biotic and abiotic risks were considered threats more often in the species-specific studies than in the others. The abiotic risks considered as threats included various phenomena such as fires, pollution, soil erosion, landslides, etc. The main biotic risk reported in the reviewed papers were invasive species.

This variability in threat definition necessitates further analysis. Urban areas are the most consistently included threat, yet they are treated as a single, undifferentiated category in over half of the studies (54%), while other studies split them into as many as seven subcategories. This inconsistency points to a lack of consensus on how much detail a threat category should include, and it limits the comparability of habitat quality outputs across studies. Because threat weights and decay distances in InVEST HQ are assigned by expert judgement rather than derived from empirical impact data, this inconsistency in threat granularity is not merely descriptive, it directly affects how much of the observed variation in habitat quality outcomes reflect genuine ecological differences between case studies, as opposed to differences in how threats were op-

erationalised. This concern is particularly relevant for HWI-related applications, where the threats most likely to drive conflicts may be masked within broader, undifferentiated threat categories.

Appropriate information on sensitivity was available in 54 of the reviewed papers; therefore, the analyses were based on data from these papers. The number of LULC classes used for habitat quality evaluation varied from three to 45 (Fig. 6). LULC with six classes was the most common, occurring in nine of the reviewed papers. The common number of LULC classes ranges between five to ten (67% of the papers), while very detailed classifications with more than 16 classes are quite rare. The average number of classes is 11.1, but it is considerably higher in the species-specific papers (15.8) than in the general papers (9.8). The habitat suitability of the LULC classes was predominantly defined as continuous values between 0 and 1. Only one paper presents a study in which habitat suitability is defined using binary values.

The near-universal reliance on continuous, expert-assigned suitability and sensitivity values, combined with the fact that appropriate sensitivity information was available in only 54 of the 61 reviewed papers, points to a broader validation gap in the InVEST HQ literature. While continuous values offer greater flexibility for differentiating habitat suitability than binary classifications, their derivation is rarely documented in sufficient detail to allow assessment of reproducibility, and cross-study comparison is further complicated by the wide range in the number of LULC classes used (3 to 45, with a mean of 11.1). This variation is even more pronounced in species-specific studies (mean of 15.8 classes) than in general applications (mean of 9.8), which is expected given the more detailed habitat requirements of individual species, but also increases the number of expert-based sensitivity judgements required per study and, consequently, the potential for subjective bias in the resulting habitat quality outputs.

The main model outputs are presented as two grids: habitat suitability and habitat degradation. Analyses of the outputs used in the reviewed papers showed that all of them use the habitat quality grid, and over one-third also incorporate the habitat degradation grid. In other words, 38 studies (62%) use the habitat quality grid in their results and 23 (38%) use both grids. The comparison

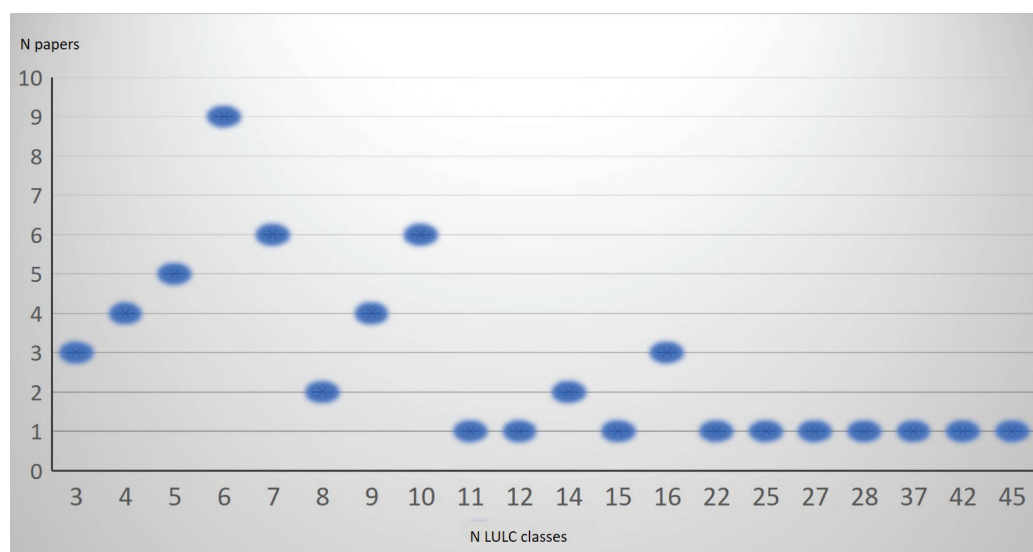


Figure 6. Number of LULC classes used for habitat valuation and the number of papers in which each of them is found.

between general and species-specific papers did not reveal any difference from the overall results.

Relation to HWI was identified in 22 of the reviewed papers. Ten of these were considered directly related, and all are species-specific (5 dealing with animal groups and 4 with specific species). The majority of these studies (6) examine the interaction between specific human activities and habitat quality of a particular species. Examples include the effect of deforestation on Tiger habitat quality (Louis et al. 2024), land cultivation, the presence of residents and traffic leading to landscape fragmentation of bird habitats (Yu et al. 2020), and the effect of human induced environmental degradation on endangered animal species (Meza Mori et al. 2025). The other aspect of HWI defined in these papers is wildlife protection under threat, covering topics such as the evaluation of protected areas designated and managed for wildlife habitat protection (Jeong et al. 2024) and the design of wildlife corridors to reduce threats to mammal species (Sulistyanawan et al. 2017). Twelve studies have an indirect relation to HWI and predominantly cover the other two aspects of HWI. Analyses of human impact on biodiversity or specific species address topics such as conflict between biodiversity conservation and economic development (Liu et al. 2022) and analyses of habitat quality impact factors in relation to human-induced climate change (Zhang et al. 2022a). Measures to ensure favourable habitats for wildlife species cover topics such as the effectiveness of corridors in supporting biodiversity conservation and restoration efforts for multiple animal species (Shen et al. 2024) and the implementation of ecosystem protection and restoration to develop sustainable agricultural practices that maintain an ecological balance between habitat quality and economic growth (Abubakar et al. 2025).

Discussion

Methodological trends and model integration

The main advantages of the InVEST HQ model—its suitability for data-scarce contexts, its capacity to resolve spatial incongruities between nature conservation and other land-use activities, its usefulness for localising planning priorities, and its applicability to scenario-based analyses — help explain the methodological trends identified in this review. One of the most frequently highlighted advantages is its suitability for data-scarce environments; the model eliminates the need for extensive field surveys and relies on easily accessible input data, making it applicable in contexts with poor data availability and budgetary constraints (Aneseyee et al. 2020; Gu et al. 2023). Secondly, the model has proven effective in addressing spatial incongruities between nature conservation and other land-use activities, particularly urban development, providing a quick, cost-effective and easy-to-use index for evaluating the impact of different planning strategies on biodiversity, even under expert-based, data-limited conditions (Di Pirro et al. 2021). Thirdly, it has demonstrated its capacity to localise planning priorities across territories and to identify key threats to biodiversity at both regional and sub-regional scales (Epifani et al. 2017). Finally, the model is well suited to scenario-based conservation planning, enabling the calculation and spatial analysis of habitat quality under different land-use

scenarios, as demonstrated in studies applying the PLUS-InVEST framework at the basin scale (He et al. 2023).

Habitat maintenance belongs to the group of ES that provide contributions necessary to sustain populations of species, that economic units ultimately use or enjoy, through the maintenance of habitats (UN et al. 2024). Habitat quality is a crucial indicator for habitat maintenance, and InVEST HQ offers a suitable approach for spatially explicit modelling of this service. A first methodological trend identified by our review is the application of the model to assess habitat quality in relation to nature conservation and its interaction with other human activities. The habitat quality assessment by InVEST HQ appears to be particularly effective for solving spatial discrepancies between nature conservation and other activities such as urban development (Di Pirro et al. 2021) or watershed management (Admasu et al. 2023). Therefore, this type of habitat quality assessment can be considered particularly useful for resolving spatial conflicts between biodiversity conservation and other human activities, as it identifies priority areas for protection, restoration, and development across urban, watershed, and forested landscapes (Jiang et al. 2024; Parvar and Salmanmahiny 2025). Although most studies in the reviewed papers did not apply a specific ES assessment, habitat quality studies provide a valuable contribution to understanding the habitat maintenance service.

A second methodological trend involves applying the model in combination with other models and methods. This approach can provide more robust and ecologically meaningful results, compensating for its reliance on expert-based parameters (Kim et al. 2026). InVEST HQ may be most effective when combined with complementary modelling approaches that address its key limitations, particularly the lack of species-specific and spatial configuration information. Integration with statistically-based species distribution models such as MaxEnt allows habitat suitability to be derived from empirical occurrence data, thereby reducing reliance on expert-based parameters and improving species-level relevance. In parallel, coupling the model with landscape ecology tools such as FRAGSTATS enables the incorporation of spatial structure, fragmentation, and connectivity metrics.

General versus species-specific applications

Beyond these methodological trends in application context and model integration, the reviewed literature also reveals a distinct pattern regarding the object of assessment. The InVEST HQ model is often considered more suitable for assessing biodiversity in general, as it relies on LULC data and common threats, and does not account for the specific needs of individual species (Terrado et al. 2016; Li et al. 2021; Broquet et al. 2024). However, our review also identified a parallel trend of using the model in species-specific studies, which accounted for almost one-third of the reviewed papers. These include studies on Tiger habitat quality in Malaysia (Louis et al. 2024), the identification of priority areas for three endangered mammal species in the Peruvian Andes (Meza Mori et al. 2025), and the impact of land-use change on habitat quality for Crocodile Lizards in China (Zhang et al. 2022b).

The analyses showed that the model can be conditionally adapted for species-specific applications by tailoring land-use classes and sensitivity parameters to the ecological requirements of a given species. However, its outputs

should be interpreted cautiously, as they represent a proxy for habitat suitability rather than a direct measure of species presence or distribution.

Relevance to human-wildlife interactions

Since species-specific applications are the primary pathway through which InVEST HQ engages with individual wildlife populations, this subsection examines in more detail how, and to what extent, this engagement extends to human-wildlife interactions. Although HWI constitute a central motivation for this review, only 22 of the 61 reviewed papers were found to have a direct or indirect relation to HWI, underscoring that this remains a comparatively underexplored application context for InVEST HQ. Of these, only ten papers established a direct relation, and notably all of them were species-specific, suggesting that engagement with HWI in the InVEST HQ literature occurs almost exclusively through species-level rather than general biodiversity applications. This pattern indicates that the model's relevance to HWI is currently realised mainly as a by-product of species-specific habitat assessments rather than through studies explicitly designed around HWI research questions.

The characteristics of the directly related studies reveal a narrow methodological profile. Most address the interaction between a specific human activity (e.g., deforestation, land cultivation, traffic) and the habitat quality of a given species (Yu et al. 2020; Louis et al. 2024; Meza Mori et al. 2025), or the evaluation and design of protected areas and wildlife corridors (Sulistyawan et al. 2017; Jeong et al. 2024). In these applications, InVEST HQ is typically used to generate a spatially explicit baseline of habitat suitability against which a specific threat or planning intervention is assessed, rather than to model the interaction itself (e.g., conflict frequency, encounter risk, or bidirectional impacts between wildlife and human activities). This points to an important methodological gap; even in the subset of studies most closely aligned with HWI, the model's threat and sensitivity parameters are calibrated to represent generic land-use pressures rather than parameters specifically capturing conflict or coexistence dynamics.

A further limitation concerns parameterisation. As in the wider dataset, species-specific HWI-related studies rely predominantly on expert-based definitions of threat weights and species sensitivity values, which are rarely validated against independent field or occurrence data. This is a particularly consequential limitation in the HWI context, since the ecological plausibility of habitat quality outputs used to infer conflict-relevant patterns depends directly on the accuracy of these expert judgements. Studies combining InVEST HQ with species distribution models such as MaxEnt offer a partial solution by grounding sensitivity parameters in empirical occurrence data, but such combinations remain rare among studies with a direct HWI focus.

The 12 papers with an indirect relation to HWI illustrate a broader, but more diffuse set of applications, including analyses of human impacts on biodiversity in relation to economic development or climate change (Liu et al. 2022; Zhang et al. 2022a), and scenario-based evaluations of corridors and restoration measures intended to safeguard favourable habitats (Shen et al. 2024; Abubakar et al. 2025). These studies demonstrate the potential of the model to inform HWI-relevant planning, but generally treat habitat quality as one input among several in wider land-use or restoration assessments, rather than as a tool purpose-built for HWI risk assessment.

Taken together, these findings indicate that a key knowledge gap remains; the translation of InVEST HQ's spatially explicit habitat quality outputs into information that directly supports HWI risk assessment and prediction (e.g., conflict hotspot mapping, encounter probability, or coexistence potential) is still largely undeveloped. Addressing this gap will likely require closer integration with methods explicitly designed to capture HWI dynamics, such as spatial conflict-risk modelling or agent-based approaches, in addition to the landscape ecology and statistical model combinations already common in the general InVEST HQ literature.

Conclusions

The results of this review highlight the increasing popularity of the InVEST HQ model as a spatially explicit tool for assessing habitat quality for nature conservation across a wide range of contexts. The model is predominantly used for general-purpose assessments of ES and land-use planning, reflecting its suitability for broad-scale analyses of trade-offs between biodiversity conservation and human activities. Although a growing body of literature demonstrates its application in species-specific studies, these remain comparatively few and thus an important research gap remains. Direct applications of InVEST HQ to HWI are relatively rare in the reviewed literature, and where such a relation exists, it is most often addressed through species-specific studies rather than explicit HWI-oriented research designs. A key challenge remains the development of methods that effectively translate spatial patterns of HWI into actionable information for risk assessment and prediction.

At the same time, the model is commonly used in combination with other modelling approaches, indicating both its flexibility and its inherent methodological limitations. Its widespread integration with complementary tools suggests that it is particularly useful as a core spatial framework, which then requires additional modelling layers to capture ecological complexity. This aligns with current trends in integrated ES modelling, which has become necessary due to the multiple trade-offs among ES and the associated challenges that ES assessments aim to address (Girardello et al. 2019; Huang et al. 2025). In this context, it is important to define more clearly which spatial patterns characterise HWI and the extent to which these patterns can be modelled for risk assessment and prediction using relatively simple and user-friendly tools. Our review suggests that such understanding is beginning to emerge but remains fragmented, largely because of the limited number of robust species-specific applications.

The diversity of model combinations identified in this review highlights the absence of a single dominant methodological framework, instead revealing a flexible, problem-driven selection of tools. The classification proposed in this study systematises this variability into a coherent structure that can support future applications and facilitate methodological comparison in habitat quality and ES assessments. Among the identified combinations, landscape ecology models (e.g., FRAGSTATS and MCR) and statistical-based approaches, such as MaxEnt, emerge as the most consistently applied, reflecting their complementary roles in capturing spatial configurations and species-environment relationships. These patterns suggest that integrative approaches combining spatial, statistical, and process-oriented methods represent the most frequently re-

ported direction in the reviewed literature, although this review did not evaluate the comparative performance or validation status of individual model combinations, thus their actual effectiveness remains to be systematically tested. This uncertainty is compounded by the reliance of InVEST HQ on expert-based parameterisation, particularly in the definition of threat weights and species sensitivity values, which are rarely validated against independent field data in the reviewed studies. At the same time, the relatively high variability and lower methodological density in species-specific studies indicates that these combinations, as well as the underlying parameterisation choices, require further testing and standardisation to support more robust and transferable applications.

Finally, we argue that future research should pursue two complementary directions. First, systematically testing and validating combined modelling frameworks that integrate InVEST HQ with species distribution models and landscape ecology approaches would help bridge the gap between general ES assessments and species-specific ecological processes, and reduce the current reliance on unvalidated expert-based parameterisation. Second, and more specifically for HWI-oriented research, future studies should move beyond using InVEST HQ solely to generate a habitat quality baseline, and instead explore its closer integration with methods explicitly designed to capture HWI dynamics, such as spatial conflict-risk modelling or agent-based approaches. A systematic assessment of how, and how often InVEST HQ applications validate their outputs – for example against independent field observations, species occurrence records, or expert review – was beyond the scope of the present review but represents an important direction for future syntheses, given the model's reliance on expert-based parameterisation identified throughout this study. Together, these directions would improve the transferability of model outputs into decision-relevant information for conservation planning and, more specifically, for human-wildlife interaction risk assessment and prediction.

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Additional information

Conflict of interest

The authors have declared that no competing interests exist.

Ethical statement

No ethical statement was reported.

Artificial Intelligence (AI) use

The authors accept full responsibility for the content of the manuscript, including the disclosure of any use of AI.

AI was used only for language polishing; all scientific content, analysis, and conclusions are the authors' own.

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Author contributions

Conceptualization: SN. Data curation: SN, KP, SY, IV, MP. Formal analysis: SN, KR, SY, IV. Investigation: SN, KP, SY, IV, MP. Methodology: SN. Project administration: SN.

Author ORCIDs

S. Nedkov  <https://orcid.org/0000-0002-0052-9815>
K. Parvanova  <https://orcid.org/0009-0003-5445-426X>
S. Yakovleva  <https://orcid.org/0009-0002-3946-6927>
I. Vasileva  <https://orcid.org/0009-0008-9669-1091>
M. Paunova  <https://orcid.org/0009-0000-3285-6653>

Data availability

All of the data that support the findings of this study are available in the main text or Supplementary Information.

Supplementary material 1

List of the reviewed papers

Author: Stoyan Nedkov

Data type: docx

Explanation note: The file contains bibliographic information for all 61 papers used in the study.

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Supplementary material 2

Review template

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Data type: typeFile

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